kVA to Amps Chart

Full-load current for the standard transformer sizes at the common voltages, three-phase and single-phase.

Three-phase amps are kVA times 1,000, divided by 1.732 times the line voltage. A 75 kVA transformer delivers 208 A at 208 V and 90 A at 480 V. There is no power factor in the conversion: kVA is already volts times amps.

Three-phase

kVA	208 V	240 V	480 V	600 V
15	41.6	36.1	18.0	14.4
30	83.3	72.2	36.1	28.9
45	125	108	54.1	43.3
75	208	180	90.2	72.2
112.5	312	271	135	108
150	416	361	180	144
225	625	541	271	217
300	833	722	361	289
500	1,388	1,203	601	481
750	2,082	1,804	902	722
1,000	2,776	2,406	1,203	962
1,500	4,164	3,608	1,804	1,443
2,000	5,551	4,811	2,406	1,925
2,500	6,939	6,014	3,007	2,406
3,000	8,327	7,217	3,608	2,887

Full-load amps per phase at the stated line-to-line voltage.

Single-phase

kVA	120 V	240 V	480 V
1	8.33	4.17	2.08
3	25.0	12.5	6.25
5	41.7	20.8	10.4
10	83.3	41.7	20.8
15	125	62.5	31.3
25	208	104	52.1
37.5	313	156	78.1
50	417	208	104
75	625	313	156
100	833	417	208
167	1,392	696	348

Full-load amps at the stated voltage.

How the values are calculated

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I = kVA × 1000 / (1.732 × V)  [three-phase]
I = kVA × 1000 / V  [single-phase]
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Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

Full-load current is the starting point for sizing conductors and overcurrent protection, not the answer. NEC 450.3 sets transformer protection as a percentage of this current, conductors are sized under Article 310 with the 125 percent continuous-load factor where it applies, and the figure says nothing about inrush or the fault current the transformer can deliver.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/kva-to-amps-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.